TUTORIAL 9

Matrix Representations, Change of Basis, and Determinants

MOTIVATION AND GUIDING IDEA

Motivation. A representing matrix is a computational tool that depends on the ordered bases of the domain and codomain. Changing coordinates leaves the map unchanged but can greatly simplify repeated calculations. Always identify the input and output coordinate systems to avoid reversing a change of basis.

NOTATION AND CONVENTIONS

- For ordered bases $B = (u_1, \dots, u_n)$ of U and $C = (v_1, \dots, v_m)$ of V, the notation $[T]_C^B$ denotes an $m \times n$ matrix taking B-coordinates as input and returning C-coordinates.
- $S \circ T$ applies T first and S second; I_V is the identity map on V.
- $P_{B \leftarrow B'} = [I_V]_B^{B'}$ takes $[v]_{B'}$ to $[v]_B$. The arrow indicates the direction of coordinate conversion.
- If B is an ordered basis of $\mathbb{F}^n$ and E is the standard basis, $S_B = [\, b_1 \ \cdots \ b_n \,] = [I]_E^B$ is the basis-column matrix. Hence $[v]_E = S_B [v]_B$.
- $[T]_B$ abbreviates $[T]_B^B$ for an operator $T : V \to V$. Square matrices A, D are similar if $D = P^{-1}AP$ for an invertible P.

DEFINITION 9.1

Let U, V be finite-dimensional vector spaces over the same field, with ordered bases B, C, and let $T : U \to V$ be linear. Its representing matrix is defined by its columns:

$$[T]_C^B = \big[\, [T(u_1)]_C \ \cdots \ [T(u_n)]_C \,\big], \qquad [T(u)]_C = [T]_C^B [u]_B.$$

For another ordered basis B' of the domain, the columns of $P_{B \leftarrow B'}$ are the B-coordinate vectors of the vectors of B'. Its inverse is $P_{B' \leftarrow B}$. If $R : V \to W$ is linear and D is an ordered basis of W, then

$$[R \circ T]_D^B = [R]_D^C [T]_C^B.$$

If T is an isomorphism, $[T^{-1}]_B^C = ([T]_C^B)^{-1}$. For an operator with standard matrix A, the matrix in the ordered basis B is $S_B^{-1} A S_B$. Domain and codomain bases must match at every matrix product.

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MOTIVATION AND GUIDING IDEA

Construct a map from prescribed vector images, check compatibility with linear dependence relations, compute a representing matrix and change bases. The emphasis is on the connection between an abstract map, coordinates and matrix calculations.

9.1 Determining a map from vector images

PROPOSITION 9.1

Let V, W be vector spaces over the same field $\mathbb{F}$, let $B = (v_1, \dots, v_n)$ be an ordered basis of V, and let $w_1, \dots, w_n \in W$ be arbitrary vectors, not necessarily independent. There is a unique linear map $T : V \to W$ with $T(v_i) = w_i$ for every i. Explicitly, if $v = \sum_{i=1}^{n} a_i v_i$, then

$$T(v) = \sum_{i=1}^{n} a_i w_i.$$

The formula is well defined because basis expansions are unique. It is linear because coordinates of sums and scalar multiples are computed entry by entry. Any linear map with the prescribed images must obey this formula, proving uniqueness. In particular, between any two vector spaces over the same field there is at least one linear map: the zero map.

EXERCISE 9.1

(a) $T(1,0,0)^T = (2,0,1)^T$, $T(1,1,0)^T = (1,1,0)^T$, $T(1,0,1)^T = (0,2,1)^T$;

(b) $T(2,1,4)^T = (1,2,0)^T$, $T(1,0,2)^T = (1,-1,1)^T$, $T(0,1,0)^T = (-1,4,0)^T$.

Please decide in each case whether a linear map $T : \mathbb{R}^3 \to \mathbb{R}^3$ satisfies the requirements. If it exists, please find an explicit formula.

WORKED SOLUTION

In part (a), the three inputs form a basis, so the map exists uniquely. Computing the images of the standard basis gives the formula. In part (b), the proposed outputs do not preserve the displayed input relation, so no such map exists.

(a) $T(e_1) = (2,0,1)^T$, $T(e_2) = (1,1,0)^T - T(e_1) = (-1,1,-1)^T$,

$$T(e_3) = (0,2,1)^T - T(e_1) = (-2,2,0)^T, \qquad T(x,y,z)^T = (2x - y - 2z, y + 2z, x - y)^T.$$

(b) $(2,1,4)^T = 2(1,0,2)^T + (0,1,0)^T$,

$$2(1,-1,1)^T + (-1,4,0)^T = (1,2,2)^T \neq (1,2,0)^T.$$

MOTIVATION AND GUIDING IDEA

Existence criterion for prescribed images. The requirements $T(u_i) = w_i$ can be satisfied by a linear map if and only if, for every choice of coefficients a_i,

$$\sum_i a_i u_i = 0 \quad \Longrightarrow \quad \sum_i a_i w_i = 0.$$

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Under this condition, define a map on $U = \operatorname{span}\{u_i\}$ by $T(\sum_i a_i u_i) = \sum_i a_i w_i$. The difference of two representations is a dependence relation, so both yield the same image. Linearity follows from the definition. If the domain is finite-dimensional, extend a basis of U to a basis of V, and freely prescribe images of the extra basis vectors. The map is unique if $U = V$. If $U \neq V$ and $W \neq \{0\}$, two different images may be chosen for an extra basis vector, so the extension is not unique. If the u_i already form a basis, there are no nontrivial dependence relations, and every choice of images defines a unique linear map.

EXERCISE 9.2

$$T : \mathbb{R}^2 \to \mathbb{R}_3[X], \qquad T((1,0)^T) = 1 + 2X^2, \quad T((1,1)^T) = 4 + X + 2X^2.$$

Please find the unique linear map satisfying these requirements.

WORKED SOLUTION

The two given inputs form a basis. Linearity forces the formula, and substitution reproduces both prescribed outputs.

$$T((0,1)^T) = T((1,1)^T) - T((1,0)^T) = 3 + X,$$

$$T((x,y)^T) = x(1 + 2X^2) + y(3 + X) = (x + 3y) + yX + 2xX^2.$$

9.2 Composition of maps

RECALL AND TOOLS

- Let U, V, W be vector spaces over a field $\mathbb{F}$.
- Let $T : U \to V$ and $S : V \to W$ be linear maps.
- The map $S \circ T : U \to W$, defined for $u \in U$ by $(S \circ T)(u) = S(T(u))$, is their **composition**.
- $S \circ T$ is linear over $\mathbb{F}$.
- If both maps are surjective, their composition is surjective. If both are injective, their composition is injective. In particular, a composition of isomorphisms is an isomorphism.

EXERCISE 9.3

$$T : \mathbb{R}^2 \to M_2(\mathbb{R}), \qquad T(a,b)^T = \begin{pmatrix} 2a & a-b \\ a+b & 3b \end{pmatrix},$$

$$S : M_2(\mathbb{R}) \to \mathbb{R}_3[X], \qquad S\begin{pmatrix} q & r \\ s & t \end{pmatrix} = qX^2 + (r - 2s)X + t.$$

Please compute $S \circ T$.

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EXERCISE 9.3 CONTINUED

WORKED SOLUTION

Apply T first and substitute its four matrix entries into the formula for S.

$$(S \circ T)(a,b)^T = (2a)X^2 + \big((a-b) - 2(a+b)\big)X + 3b = 2aX^2 - (a+3b)X + 3b.$$

9.3 Representing a linear map by a matrix

DEFINITION 9.2

Let U be an n-dimensional vector space over $\mathbb{F}$ with ordered basis $B = (u_1, \ldots, u_n)$.
Let V be an m-dimensional vector space over the same field with ordered basis $C = (v_1, \ldots, v_m)$.
For a linear map $T : U \to V$, define the matrix

$$[T]_C^B = \begin{pmatrix} | & | & | & | \\ [T(u_1)]_C & [T(u_2)]_C & \cdots & [T(u_n)]_C \\ | & | & | & | \end{pmatrix} \in \mathbb{F}^{m \times n}$$

$[T]_C^B$ is the **matrix of T relative to the ordered bases B and C.**

Construction

First compute $T(u_i)$, the images of the basis vectors of the domain.

Then express each image in coordinates relative to C, and use these coordinate columns in the given order.

PROPOSITION 9.2

Let U, V be vector spaces over $\mathbb{F}$, with ordered bases B of U and C of V. For a linear map $T : U \to V$ and every $u \in U$,

$$[T(u)]_C = [T]_C^B \cdot [u]_B$$

Compute this matrix once for the chosen ordered bases. It then converts the coordinate vector of any new input into the coordinate vector of its image relative to C.

EXERCISE 9.4

$$V = M_3(\mathbb{R}), \qquad T(A) = 2(A - A^T),$$
$$U = \{A : A = A^T\}, \qquad W = \{A : A = -A^T\}.$$

Please prove linearity, please find the kernel and image and their dimensions, and please give a representing matrix in a basis formed by a basis of U followed by a basis of W.

WORKED SOLUTION

Here E_{ij} has a single 1 in position (i,j) and zeros elsewhere. The free matrix entries give the displayed bases. The map vanishes on symmetric matrices and multiplies skew-symmetric

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EXERCISE 9.4 CONTINUED

matrices by 4, giving the diagonal representation.

$$T(\alpha A+\beta B)=2\big(\alpha A+\beta B-\alpha A^T-\beta B^T\big)=\alpha T(A)+\beta T(B).$$

$$T(A)=0 \iff A=A^T, \qquad \ker T=U.$$

$$T(A)^T=-T(A), \qquad K\in W \implies T(K/4)=K.$$

$$\operatorname{Im} T=W, \qquad B_U=(E_{11},E_{22},E_{33},E_{12}+E_{21},E_{13}+E_{31},E_{23}+E_{32}),$$

$$B_W=(E_{12}-E_{21},E_{13}-E_{31},E_{23}-E_{32}), \qquad \dim U=6, \quad \dim W=3, \quad 6+3=9.$$

$$A=\frac{A+A^T}{2}+\frac{A-A^T}{2}\in U+W, \quad U\cap W=\{0\}, \qquad [T]_{(B_U,B_W)}=\operatorname{diag}(0,0,0,0,0,0,4,4,4).$$

EXERCISE 9.5

$$[T]_E=A=\begin{pmatrix}0&2&-1\\0&3&1\\0&0&4\end{pmatrix}, \qquad V=\mathbb{R}^3.$$

Please find nonzero vectors v,u satisfying $T(v)=0$ and $T(u)=4u$. Then please find w satisfying $T(w)=w+u$.

WORKED SOLUTION

The first column is zero, so e_1 lies in the kernel. Solving the second system gives an eigenvector for 4. Scaling that eigenvector by $\frac{1}{3}$ yields the last equation.

$$v=(1,0,0)^T, \qquad Av=0.$$

$$(A-4I)u=0 \iff -4u_1+2u_2-u_3=0, \quad -u_2+u_3=0.$$

$$u=(1,4,4)^T, \qquad Au=(4,16,16)^T=4u.$$

$$w=\tfrac{1}{3}u=(\tfrac{1}{3},\tfrac{4}{3},\tfrac{4}{3})^T, \qquad T(w)=\tfrac{4}{3}u=w+u.$$

EXERCISE 9.6

Let $V=\mathbb{C}[x]$, the vector space of all complex polynomials.
Define the linear operator $N:V\to V$ by

$$(Np)(x):=-(7-x)\,p'(x).$$

For each $\theta\in\mathbb{R}$, define $U_\theta:V\to V$ by

$$U_\theta:=\exp(-i\theta N^2)=\sum_{k=0}^{\infty}\frac{(-i\theta)^k}{k!}N^{2k}.$$

Please use the following two ordered bases:

$$E=(1,x,x^2,x^3,\dots), \qquad B=(7,\ (7-x),\ (7-x)^2,\ (7-x)^3,\dots).$$

1. Please prove that N and U_θ are linear.

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EXERCISE 9.6 CONTINUED

2. For $n \geq 1$, set $b_n(x) = (7-x)^n$, and set $b_0(x) = 7$. Please compute $N(b_n)$ and $N^2(b_n)$.
3. Please compute $U_\theta(b_n)$ for all $n \geq 0$ and please deduce the form of $[U_\theta]_B$.
4. Please find the matrix of U_θ restricted to $P_3 = \text{span}\{1, x, x^2, x^3\}$ in the ordered basis $E_3 = (1, x, x^2, x^3)$.

WORKED SOLUTION

1. Differentiation is linear, so

$$N(\alpha p + \beta r) = -(7-x)(\alpha p' + \beta r') = \alpha N(p) + \beta N(r).$$

To justify the series, restrict to P_d, the polynomials of degree at most d. This space is invariant under N, so on P_d the series is the exponential series of a finite matrix, converging for every θ. These definitions agree as d increases and therefore define a map on $\mathbb{C}[x]$. Linearity follows by taking limits of the partial sums in each coordinate.

2. For $n \geq 1$,

$$b_n'(x) = \frac{d}{dx}(7-x)^n = n(7-x)^{n-1} \cdot (-1) = -n(7-x)^{n-1}.$$

so

$$N(b_n) = -(7-x)\, b_n'(x) = -(7-x)\big(-n(7-x)^{n-1}\big) = n(7-x)^n = nb_n.$$

and hence

$$N^2(b_n) = N(nb_n) = nN(b_n) = n \cdot nb_n = n^2 b_n.$$

Also $N(b_0) = N(7) = 0$, so $N^2(b_0) = 0$.

3. Since $N^{2k}(b_n) = n^{2k} b_n$ for $n \geq 1$, the series gives

$$U_\theta(b_n) = \sum_{k=0}^{\infty} \frac{(-i\theta)^k}{k!}\, n^{2k}\, b_n = \exp(-i\theta n^2)\, b_n.$$

For $n = 0$, $U_\theta(b_0) = b_0$.

Thus the matrix in B is diagonal:

$$[U_\theta]_B = \text{diag}\big(1,\ e^{-i\theta},\ e^{-i4\theta},\ e^{-i9\theta}, \dots\big).$$

4. Write $q_1 = e^{-i\theta}$, $q_4 = e^{-i4\theta}$ and $q_9 = e^{-i9\theta}$. On P_3, use b_0, b_1, b_2, b_3. Their images are $b_0, q_1 b_1, q_4 b_2, q_9 b_3$, respectively.

Expand these four polynomials in E_3:

$$b_0 = 7, \qquad b_1 = 7 - x, \qquad b_2 = (7-x)^2 = x^2 - 14x + 49,$$

$$b_3 = (7-x)^3 = -x^3 + 21x^2 - 147x + 343.$$

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EXERCISE 9.6 CONTINUED

Let C be the matrix with coordinate columns $[b_0]_{E_3}, [b_1]_{E_3}, [b_2]_{E_3}, [b_3]_{E_3}$:

$$C = \begin{pmatrix} 7 & 7 & 49 & 343 \\ 0 & -1 & -14 & -147 \\ 0 & 0 & 1 & 21 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

This matrix is invertible. The formula $[U_\theta]_{E_3} = C\,\mathrm{diag}(1, q_1, q_4, q_9)C^{-1}$ gives the following matrix:

$$[U_\theta]_{E_3} = \begin{pmatrix} 1 & 7(1-q_1) & 49(1-2q_1+q_4) & 343(1-3q_1+3q_4-q_9) \\ 0 & q_1 & 14(q_1-q_4) & 147(q_1-2q_4+q_9) \\ 0 & 0 & q_4 & 21(q_4-q_9) \\ 0 & 0 & 0 & q_9 \end{pmatrix}.$$

EXERCISE 9.7

$$T : \mathbb{R}^3 \to \mathbb{R}^2, \qquad T(x,y,z)^T = (3x+2y-z, 2x-y+4z)^T.$$

Please compute the standard-basis matrix using the images of the basis vectors.

WORKED SOLUTION

The basis-vector images are the matrix columns. The final multiplication verifies the formula for every input.

$$T(e_1) = (3,2)^T, \quad T(e_2) = (2,-1)^T, \quad T(e_3) = (-1,4)^T,$$

$$[T] = \begin{pmatrix} 3 & 2 & -1 \\ 2 & -1 & 4 \end{pmatrix}, \qquad [T]\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3x+2y-z \\ 2x-y+4z \end{pmatrix}.$$

EXERCISE 9.8

$$T : \mathbb{R}_2[X] \to \mathbb{R}^2, \qquad T(p) = (p(0), p(-1))^T,$$
$$B = (1, 2+X, 1+X+2X^2), \qquad C = ((1,-1)^T, (0,1)^T).$$

Please compute $[T]_C^B$ and please verify $[T(p)]_C = [T]_C^B [p]_B$ for $p = a + bX + cX^2$.

WORKED SOLUTION

Compute each image and convert it to C-coordinates. To verify the identity, first equate coefficients in the B-expansion and then multiply by the matrix.

$$[(r,s)^T]_C = (r, r+s)^T,$$

$$T(1) = (1,1)^T, \quad T(2+X) = (2,1)^T, \quad T(1+X+2X^2) = (1,2)^T,$$

$$M = [T]_C^B = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 3 \end{pmatrix}.$$

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EXERCISE 9.8 CONTINUED

$$p = \alpha + \beta(2+X) + \gamma(1+X+2X^2) \implies \gamma = c/2, \quad \beta = b - c/2, \quad \alpha = a - 2b + c/2.$$

$$[p]_B = \begin{pmatrix} a - 2b + c/2 \\ b - c/2 \\ c/2 \end{pmatrix}, \qquad M[p]_B = \begin{pmatrix} a \\ 2a - b + c \end{pmatrix}.$$

$$T(p) = (a, a - b + c)^T, \qquad [T(p)]_C = (a, 2a - b + c)^T.$$

9.3.1 Change-of-coordinate matrices

RECALL AND TOOLS

Definition: Let V be a finite-dimensional vector space over $\mathbb{F}$, with ordered bases B, B'. The **change-of-coordinate matrix** $[I_V]_B^{B'}$ from B' to B is the representing matrix $[T]_B^{B'}$ for the **identity map** $T : V \to V$, $T(v) = v$ for every $v \in V$.

- The fundamental identity is $[v]_B = [I_V]_B^{B'}[v]_{B'}$.
- The identity map is denoted I_V, with $I_V(v) = v$.
- Its columns are the coordinate vectors of the vectors of B' relative to the ordered basis B.

EXERCISE 9.9

$$B = (1+X, 2+3X), \qquad B' = (3+4X, 1+2X), \qquad p = a + bX.$$

Please compute the transition matrix $[I]_B^{B'}$ and please verify $[p]_B = [I]_B^{B'}[p]_{B'}$.

WORKED SOLUTION

The basis matrices contain polynomial coefficients in $(1, X)$. The transition first converts to standard coordinates and then to B. The coordinate formulas and final product explicitly verify the identity.

$$S_B = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}, \quad S_{B'} = \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}, \quad S_B^{-1} = \begin{pmatrix} 3 & -2 \\ -1 & 1 \end{pmatrix},$$

$$[I]_B^{B'} = S_B^{-1} S_{B'} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

$$[p]_B = \begin{pmatrix} 3a - 2b \\ -a + b \end{pmatrix}, \qquad [p]_{B'} = \begin{pmatrix} a - b/2 \\ -2a + 3b/2 \end{pmatrix},$$

$$\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a - b/2 \\ -2a + 3b/2 \end{pmatrix} = \begin{pmatrix} 3a - 2b \\ -a + b \end{pmatrix}.$$

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9.3.2 Matrices of compositions and inverse maps

PROPOSITION 9.3

Let U, V, W be finite-dimensional vector spaces over $\mathbb{F}$, with ordered bases A, B, C, respectively.
Let $S : U \to V$ and $T : V \to W$ be linear. Then

$$[T \circ S]_C^A = [T]_C^B [S]_B^A$$

Corollary

If $T : V \to W$ is an **isomorphism** and B, C are ordered bases of V, W, respectively, then

$$[T^{-1}]_B^C = \left([T]_C^B\right)^{-1}$$

Additional practice

EXERCISE 9.10 FOUNDATION

$$T(x, y) = (2x - y, x + y), \quad B = ((1, 0)^T, (1, 1)^T), \quad C = ((1, 1)^T, (0, 1)^T), \quad v = (3, 2)^T.$$

Please compute $[T]_C^B$ and $[T^{-1}]_B^C$. Please verify the coordinate identity and please recover v using the inverse.

WORKED SOLUTION

Convert the images of the B-vectors to C-coordinates and place them in the columns. The inverse takes C-coordinates and returns B-coordinates. Reconstruction in B recovers $(3, 2)^T$.

$$[(r, s)^T]_C = (r, s - r)^T, \quad T(1, 0)^T = (2, 1)^T, \quad T(1, 1)^T = (1, 2)^T.$$

$$M = [T]_C^B = \begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}, \quad \det M = 3, \quad [T^{-1}]_B^C = M^{-1} = \frac{1}{3}\begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}.$$

$$v = (1, 0)^T + 2(1, 1)^T, \quad [v]_B = (1, 2)^T, \quad T(v) = (4, 5)^T,$$

$$[T(v)]_C = (4, 1)^T = M(1, 2)^T, \qquad M^{-1}(4, 1)^T = \frac{1}{3}(3, 6)^T = (1, 2)^T.$$

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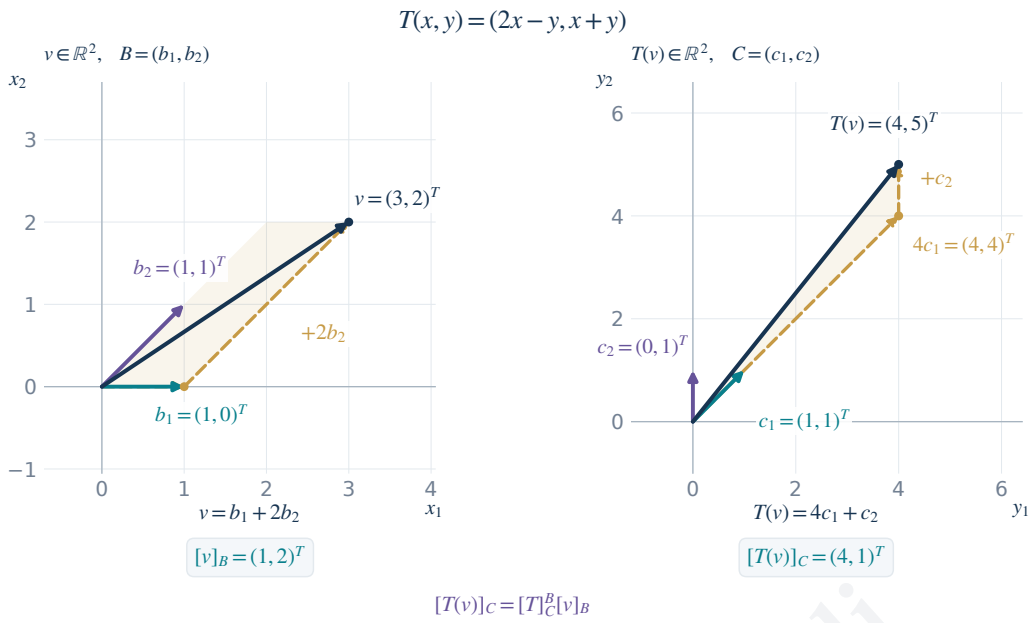


Figure 9.1. The arrows show vectors in standard coordinates; the gold paths show the decompositions $v = b_1 + 2b_2$ and $T(v) = 4c_1 + c_2$. Coordinate columns appear separately: $[v]_B = (1,2)^T$ and $[T(v)]_C = (4,1)^T$. The vector $T(v)$ itself is $(4,5)^T$.

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EXERCISE 9.11 CHALLENGE

$$A = \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix}, \quad B = ((1,1)^T, (1,-1)^T), \quad x_{k+1} = Ax_k, \quad x_0 = (2,1)^T.$$

Please compute $D = [T]_B$ for $T(x) = Ax$ and $P = [I]_E^B$. Please prove $A^k = PD^kP^{-1}$ for every $k \geq 0$ and please solve the recurrence.

WORKED SOLUTION

The columns of P are the new basis vectors in standard coordinates. Cancelling adjacent $P^{-1}P$ factors proves the power identity; for $k = 0$ both sides are the identity. Induction on the recurrence gives $x_k = A^k x_0$. In the basis B, the two components are multiplied separately by 6 and 2.

$$A(1,1)^T = 6(1,1)^T, \quad A(1,-1)^T = 2(1,-1)^T,$$

$$D = \begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}, \quad P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad P^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad A = PDP^{-1},$$

$$A^k = PD^kP^{-1} = \frac{1}{2}\begin{pmatrix} 6^k + 2^k & 6^k - 2^k \\ 6^k - 2^k & 6^k + 2^k \end{pmatrix},$$

$$x_k = A^k x_0 = \frac{1}{2}\begin{pmatrix} 3\cdot 6^k + 2^k \\ 3\cdot 6^k - 2^k \end{pmatrix}, \quad x_0 = \frac{1}{2}(4,2)^T = (2,1)^T.$$

MOTIVATION AND GUIDING IDEA

By the end of this tutorial, you will be able to choose an efficient determinant calculation, track the effect of row operations, identify conditions for invertibility, and connect algebraic structure with changes of basis and area.

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9.4 The Determinant

The determinant is defined for square matrices. In the real plane, its absolute value is the area scale factor; its sign indicates whether orientation is preserved or reversed.

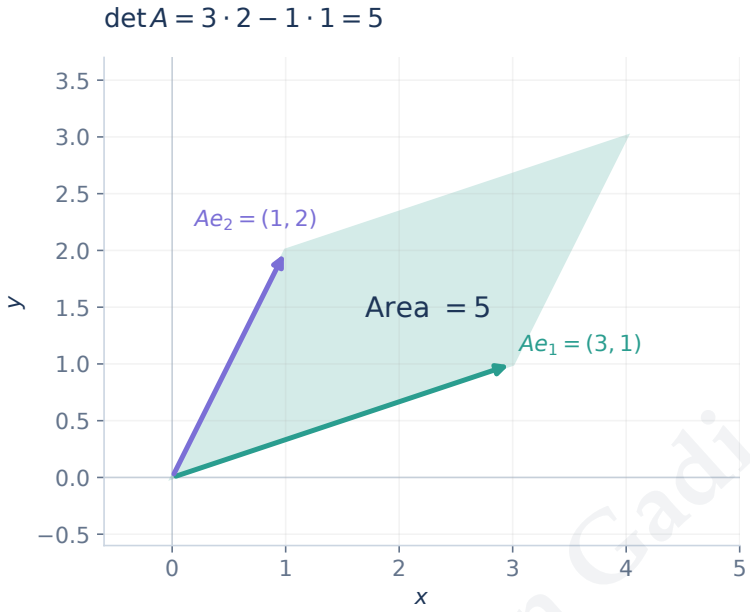


Figure 9.2. A linear map sends the unit square to a parallelogram. The parallelogram's area is the absolute value of the determinant.

DEFINITION 9.3

Let $n \in \mathbb{N}$. A function

$$\det : M_n(\mathbb{R}) \longrightarrow \mathbb{R}$$

is called the **determinant** if, for every $A \in M_n(\mathbb{R})$, it has the following properties:

1. **Interchanging rows (change of sign):** For $1 \leq i < j \leq n$, if $E^{(1)}_{i,j} \in M_n(\mathbb{R})$ is the elementary matrix interchanging rows i and j, then
$$\det\left(E^{(1)}_{i,j} A\right) = -\det(A).$$

2. **Multiplying a row by a scalar (homogeneity):** For $1 \leq i \leq n$ and $\lambda \in \mathbb{R} \setminus \{0\}$, if $E^{(2)}_{i,\lambda} \in M_n(\mathbb{R})$ is the elementary matrix multiplying row i by λ, then
$$\det\left(E^{(2)}_{i,\lambda} A\right) = \lambda \det(A).$$

3. **Adding a multiple of another row (invariance):** For $1 \leq i \neq j \leq n$ and $\lambda \in \mathbb{R}$, if $E^{(3)}_{i,j,\lambda} \in M_n(\mathbb{R})$ is the elementary matrix performing $R_i \leftarrow R_i + \lambda R_j$, then
$$\det\left(E^{(3)}_{i,j,\lambda} A\right) = \det(A).$$

4. **Normalization:**
$$\det(I_n) = 1.$$

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RECALL AND TOOLS

Two useful basic properties:

- If $A = (a_{ij}) \in M_n(\mathbb{R})$ is upper triangular, meaning that $a_{ij} = 0$ for every $i > j$, then

$$\det(A) = \prod_{k=1}^{n} a_{kk}.$$

- If rows i and j are equal for some $i \neq j$, or if A has a zero row, then $\det(A) = 0$. Similarly, two equal columns or a zero column imply $\det(A) = 0$.

EXERCISE **9.12**

Please compute the determinants using row operations, stating the effect of each operation on the determinant.

1. For $t \in \mathbb{R}$,

$$A(t) = \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 2 & 1 & t & 9 \end{pmatrix}.$$

2.

$$B = \begin{pmatrix} 2 & 1 & 0 & 3 & -1 \\ 0 & 1 & 2 & -1 & 1 \\ 2 & -2 & -6 & 6 & -4 \\ 1 & 0 & 1 & 0 & 2 \\ 0 & 2 & 0 & 1 & 1 \end{pmatrix}.$$

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WORKED SOLUTION

1. Add multiples of other rows to the fourth row; these operations preserve the determinant:

$$A(t) \xrightarrow{R_4 \leftarrow R_4 - 2R_1} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & -1 & t & 11 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 + R_2} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & t+2 & 12 \end{pmatrix}$$

$$\xrightarrow{R_4 \leftarrow R_4 - (t+2)R_3} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 8-2t \end{pmatrix}.$$

Multiplying the diagonal entries gives $\det A(t) = 1 \cdot 1 \cdot 1 \cdot (8 - 2t) = 8 - 2t$.

2. The rows of B satisfy $r_3 = r_1 - 3r_2$. Thus the two operations $R_3 \leftarrow R_3 - R_1$ and $R_3 \leftarrow R_3 + 3R_2$ produce a zero row without changing the determinant. A matrix with

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EXERCISE 9.12 CONTINUED

a zero row has determinant zero, so $\det B = 0$.

RECALL AND TOOLS

Basic properties (from the lectures):

- **Multiplicativity:** For every $A, B \in M_n(\mathbb{R})$,
$$\det(AB) = \det(A)\det(B).$$
- **Failure of additivity (in general):** In general, the following equality does not hold:
$$\det(A+B) = \det(A) + \det(B).$$
- **Invariance under transposition:** For every $A \in M_n(\mathbb{R})$,
$$\det(A^T) = \det(A).$$
- **Scaling the entire matrix:** For every $c \in \mathbb{R}$ and $A \in M_n(\mathbb{R})$,
$$\det(cA) = c^n \det(A),$$
where n is the size of the matrix.

Fundamental theorem:

PROPOSITION 9.4

For every $A \in M_n(\mathbb{R})$, A is invertible if and only if $\det(A) \neq 0$.

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EXERCISE 9.13

Let $A, B \in M_n(\mathbb{R})$, with A singular and B invertible. Please prove or disprove that the matrix in each part is singular:

1. $A^T B$.
2. $B^{-1}AB$.
3. $A + A^2 + A^3 + A^4$.
4. $(A+I)^2 - (A-I)^2$.
5. $(A+B)^2 + (A-B)^2$.

WORKED SOLUTION

Use $\det A = 0$, $\det B \neq 0$, and multiplicativity of the determinant.

1. $\det(A^T B) = \det(A^T)\det B = \det A \det B = 0$.
2. $\det(B^{-1}AB) = \det(B^{-1})\det A \det B = 0$.

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EXERCISE 9.13 CONTINUED

3. Factor out A on the left:

$$A + A^2 + A^3 + A^4 = A(I + A + A^2 + A^3),$$

Therefore the determinant is zero.

4. Since $AI = IA = A$, we obtain

$$(A + I)^2 - (A - I)^2 = (A^2 + 2A + I) - (A^2 - 2A + I) = 4A.$$

Consequently, $\det(4A) = 4^n \det A = 0$.

5. Expand while preserving the order of multiplication:

$$(A + B)^2 + (A - B)^2 = (A^2 + AB + BA + B^2) + (A^2 - AB - BA + B^2) = 2(A^2 + B^2).$$

The determinant is therefore $2^n \det(A^2 + B^2)$, but it need not vanish. A counterexample is $A = 0$, $B = 2I_n$: the resulting matrix is $8I_n$, which is invertible and has determinant $8^n \neq 0$.

9.5 Determinants and Change of Basis

RECALL AND TOOLS

We also use determinants over a field $\mathbb{F}$, such as $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{Z}_p$ with p prime. Arithmetic in the formulas is performed in that field. In particular,

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

If $\mathcal{B}, \mathcal{C}$ are bases of the same space and the columns of S are the coordinates of the vectors of $\mathcal{C}$ in $\mathcal{B}$, then

$$[T]_{\mathcal{C}} = S^{-1}[T]_{\mathcal{B}}S.$$

Multiplicativity gives

$$\det[T]_{\mathcal{C}} = \det(S^{-1}) \det[T]_{\mathcal{B}} \det S = \det[T]_{\mathcal{B}}.$$

Here $T : V \to V$ is linear and $\dim V < \infty$. Thus we may define $\det T = \det[T]_{\mathcal{B}}$; this scalar does not depend on the choice of basis.

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EXERCISE 9.14

Let V be a real vector space with basis $\mathcal{B} = (b_1, b_2)$, and let $T : V \to V$ satisfy

$$T(b_1) = 4b_1 + b_2, \qquad T(b_2) = b_1 + 2b_2.$$

For $\mathcal{C} = (b_1, b_1 + b_2)$, please compute $[T]_{\mathcal{B}}$ and $[T]_{\mathcal{C}}$, and please verify that their determinants agree.

WORKED SOLUTION

The images of the basis vectors form the columns, so

$$A = [T]_{\mathcal{B}} = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \det A = 4 \cdot 2 - 1 \cdot 1 = 7.$$

Write $c_1 = b_1$, $c_2 = b_1 + b_2$. Then $b_2 = c_2 - c_1$, and hence

$$T(c_1) = 4c_1 + (c_2 - c_1) = 3c_1 + c_2,$$

$$T(c_2) = T(b_1) + T(b_2) = 5b_1 + 3b_2 = 2c_1 + 3c_2.$$

Therefore

$$[T]_{\mathcal{C}} = \begin{pmatrix} 3 & 2 \\ 1 & 3 \end{pmatrix}, \qquad \det[T]_{\mathcal{C}} = 3 \cdot 3 - 2 \cdot 1 = 7.$$

The change-of-basis calculation gives the same result:

$$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad S^{-1}AS = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & 5 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 1 & 3 \end{pmatrix}.$$

EXERCISE 9.15 EXAM QUESTION

Work over $\mathbb{Z}_3 = \{0, 1, 2\}$, with all arithmetic modulo 3. For each $\omega \in \mathbb{Z}_3$, define

$$A(\omega) = \begin{pmatrix} 1 & \omega & 1 \\ \omega & 1 & 1 \\ 1 & 1 & \omega \end{pmatrix}, \qquad b = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \qquad c = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}.$$

1. Please determine the values of ω for which $b \in \mathrm{Col}(A(\omega))$, and please find all solutions of $A(\omega)x = b$ in those cases.
2. Please determine the values of ω for which both $b \in \mathrm{Col}(A(\omega))$ and $c \in \mathrm{Col}(A(\omega))$ hold.

WORKED SOLUTION

All operations below are modulo 3. For $\omega = 0, 2$ the matrix is invertible, so the displayed solutions for b are unique. For $\omega = 1$ its column space consists of equal-entry vectors, so the system with b is inconsistent. Both parts therefore have the answer $\omega \in \{0, 2\}$. The last products also give c in each of these two cases.

$$A(0) = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad A(1) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad A(2) = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

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EXERCISE 9.15 EXAM QUESTION CONTINUED

$$\det A(\omega) = -\omega^3 + 3\omega - 2, \qquad \det A(0) = 1, \quad \det A(2) = 2 \quad \text{in } \mathbb{Z}_3.$$

$$A(0)\begin{pmatrix}0\\1\\2\end{pmatrix} = \begin{pmatrix}2\\0\\1\end{pmatrix} = b, \qquad A(2)\begin{pmatrix}0\\2\\1\end{pmatrix} = \begin{pmatrix}2\\0\\1\end{pmatrix} = b.$$

$$\operatorname{Col}(A(1)) = \{(a,a,a)^T : a \in \mathbb{Z}_3\}, \qquad b \notin \operatorname{Col}(A(1)).$$

$$A(0)\begin{pmatrix}2\\0\\2\end{pmatrix} = \begin{pmatrix}1\\2\\2\end{pmatrix} = c, \qquad A(2)\begin{pmatrix}0\\2\\0\end{pmatrix} = \begin{pmatrix}1\\2\\2\end{pmatrix} = c.$$

EXERCISE 9.16 EXAM QUESTION

Please work in $\mathbb{Z}_7$. Please solve the system in terms of $\omega \in \mathbb{Z}_7$:

$$\begin{cases} x + y + z = 2, \\ x + \omega y + z = 5, \\ x + y + (\omega + 1)z = 2. \end{cases}$$

Please write the solution set and number of solutions in every case.

WORKED SOLUTION

Subtract the first equation from each of the other two:

$$\begin{cases} x + y + z = 2, \\ (\omega - 1)y = 3, \\ \omega z = 0. \end{cases}$$

1. If $\omega = 1$, the equation $0 = 3$ gives no solution.
2. If $\omega = 0$, then $-y = 3$, so $y = 4$. Choosing $z = t$ gives

   $$S_0 = \{(5 - t, 4, t) : t \in \mathbb{Z}_7\}.$$

   There are exactly seven solutions, since different values of t give different third coordinates.
3. If $\omega \in \{2,3,4,5,6\}$, both ω and $\omega - 1$ are invertible. Thus

   $$z = 0, \qquad y = 3(\omega - 1)^{-1}, \qquad x = 2 - y.$$

   Each case has a unique solution:

ω	2	3	4	5	6
x	6	4	1	3	0
y	3	5	1	6	2
z	0	0	0	0	0

As a check, the coefficient determinant is $\omega(\omega - 1)$, which vanishes only at $\omega = 0, 1$.

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EXERCISE 9.17

$$m = Cq, \qquad C = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{pmatrix}.$$

Please find a column-space basis. Please prove that C is invertible and please calculate C^{-1} and q in terms of m.

WORKED SOLUTION

The matrix is triangular with diagonal entries 1, so it is invertible and its columns form a basis. Solving the equations in order gives the inverse formula.

$$\det C = 1, \qquad B_{\text{col}} = \{(1,2,-1)^T, (0,1,3)^T, (0,0,1)^T\}.$$

$$m_1 = q_1, \quad m_2 = 2q_1 + q_2, \quad m_3 = -q_1 + 3q_2 + q_3.$$

$$q_1 = m_1, \quad q_2 = m_2 - 2m_1, \quad q_3 = m_3 - 3m_2 + 7m_1.$$

$$C^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 7 & -3 & 1 \end{pmatrix}, \qquad CC^{-1} = C^{-1}C = I_3.$$

EXERCISE 9.18

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 \in \mathbb{F}_{\le 3}[x], \qquad \ell_1(p) = p(0), \quad \ell_2(p) = p(-1), \quad \ell_3(p) = p'(0).$$

Please find the matrix of the three maps in the basis $(1, x, x^2, x^3)$, row-space and column-space bases, and the rank. Please find all p satisfying $p(0) = 2$, $p(-1) = 3$, $p'(0) = 1$, and the dimension of the solution set.

WORKED SOLUTION

Expanding the evaluations and derivative gives the matrix. The displayed minor is invertible over every field, so the rank is 3 and the listed rows and columns are bases. The solution set is an affine line of dimension 1. Integer constants denote the corresponding field elements, so the calculation also holds in positive characteristic.

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & -1 & 1 & -1 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad \det \begin{pmatrix} 1 & 0 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & 0 \end{pmatrix} = -1.$$

$$B_{\text{row}} = \{(1,0,0,0), (1,-1,1,-1), (0,1,0,0)\}, \qquad B_{\text{col}} = \{(1,1,0)^T, (0,-1,1)^T, (0,1,0)^T\}.$$

$$a_0 = 2, \quad a_1 = 1, \quad 2 - 1 + a_2 - a_3 = 3 \implies a_2 = 2 + a_3.$$

$$p(x) = 2 + x + 2x^2 + t(x^2 + x^3), \qquad t \in \mathbb{F}.$$

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EXERCISE 9.19

$$A = \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \end{pmatrix}, \qquad z = ax_1 + bx_2 + cx_3.$$

Please find a necessary and sufficient condition for uniquely recovering $x \in \mathbb{R}^3$ from Ax and z. Please give one useful extra reading and one reading determined by the original two.

WORKED SOLUTION

Unique recovery means that the extra reading does not vanish on the kernel direction. Equivalently, the augmented square matrix is invertible. The first example satisfies the condition; the second is the sum of the two original readings.

$$\ker A = \operatorname{span}\{(3,-2,1)^T\}, \qquad (a,b,c)(3,-2,1)^T = 3a - 2b + c.$$

$$\det \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \\ a & b & c \end{pmatrix} = 3a - 2b + c.$$

$$3a - 2b + c \neq 0.$$

$$z = x_3: \quad (a,b,c) = (0,0,1), \qquad z = x_1 + x_2 - x_3: \quad (a,b,c) = (1,1,-1).$$

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